

4-EDA-noise

1 Noise Threshold Calculation

This notebook focuses on quantifying and separating measurement noise from process-driven variability in the dataset. An advantage of this dataset is the presence of six repeated measurements (SheetId) under identical printing conditions defined by (Color, Chip, Nozzle, V, F_r, dt2, Coverage). These repetitions allow for direct estimation of the noise threshold.

We will define Noise as

Noise = $\text{Var}(\text{SD} \mid \text{same (Color, Chip, Nozzle, V, F_r, dt2, Coverage)})$

```
[1]: import pandas as pd
      from pathlib import Path

      df_raw = pd.read_parquet("../data/raw/Waveform_tuning.parquet")

      col_missing = df_raw.isna().mean()
      empty_cols = col_missing[col_missing == 1.0]

      df_clean = df_raw.drop(columns=empty_cols.index)

[2]: cols_meta = ["SheetId#", "Color$", "HeadIdx#", "V", "F_r", "dt2", "Coverage#"]
      cols_values = [c for c in df_clean.columns if c.startswith("Value_")]

      df_model = df_clean[cols_meta + cols_values]
```

Testing We will start with one configuration before using the whole dataset

```
[3]: subset = df_model[
      (df_model["V"] == 20.0) &
      (df_model["F_r"] == 1.02) &
      (df_model["dt2"] == -1100.0) &
      (df_model["Coverage#"] == 2.0)
      ]
```

Melt the subset to long format:

```
[4]: long = subset.melt(
      id_vars=["SheetId#", "Color$", "HeadIdx#"],
```

```

        value_vars=cols_values,
        var_name="Nozzle",
        value_name="SD"
    )

    long["Nozzle"] = long["Nozzle"].str.extract(r'(\d+)').astype(int)

```

Calculate variance across sheets:

```

[15]: noise = (
        long.groupby(["Color$", "HeadIdx#", "Nozzle"])["SD"]
        .std()
        .reset_index(name="noise_std")
    )

    noise["noise_std"].describe()

```

```

[15]: count    134400.000000
      mean         0.001517
      std         0.000611
      min         0.000089
      25%         0.001109
      50%         0.001449
      75%         0.001845
      max         0.025632
      Name: noise_std, dtype: float64

```

Validation Validation (making sure we have 6 sheets for every configuration):

```

[6]: long.groupby(["Color$", "HeadIdx#", "Nozzle"])["SheetId#"].nunique().
      ↪value_counts()

```

```

[6]: SheetId#
     6    133280
     5     1120
     Name: count, dtype: int64

```

```

[7]: long.groupby("SheetId#").size()

```

```

[7]: SheetId#
     980301    33600
     980302    33600
     980303    33600
     980304    33600
     980305    33600
     980306    32480
     980307    33600
     980308    33600

```

```

980309    33600
980310    33600
980311    33600
980312    33600
980313    33600
980314    33600
980315    33600
980316    33600
980317    33600
980318    33600
980319    33600
980320    33600
980321    33600
980322    33600
980323    33600
980324    33600
dtype: int64

```

Sheet 980306 has incomplete nozzle coverage.

```

[8]: long = subset.melt(
      id_vars=["SheetId#", "Color$", "HeadIdx#", "V", "F_r", "dt2", "Coverage#"],
      value_vars=cols_values,
      var_name="Nozzle",
      value_name="SD"
    )

```

```

[9]: out = long.groupby(
      ["Color$", "HeadIdx#", "V", "F_r", "dt2", "Coverage#", "SheetId#"]
    ).size().unstack(fill_value=0)

    out.to_csv("debug_output.csv")

```

```

[10]: long[long["SheetId#"] != 980306]["Nozzle"].nunique()

```

```

[10]: 1120

```

```

[11]: counts = long.groupby(
      ["Color$", "HeadIdx#", "Nozzle", "V", "F_r", "dt2", "Coverage#"]
    )["SheetId#"].nunique().reset_index()

    broken = counts[counts["SheetId#"] != 6]

    broken.head(1120)

```

```

[11]:
      Color$  HeadIdx#    Nozzle    V  F_r    dt2  Coverage#  SheetId#
28000      C        26  Value_000# 20.0  1.02 -1100.0         2.0         5
28001      C        26  Value_001# 20.0  1.02 -1100.0         2.0         5

```

28002	C	26	Value_002#	20.0	1.02	-1100.0	2.0	5
28003	C	26	Value_003#	20.0	1.02	-1100.0	2.0	5
28004	C	26	Value_004#	20.0	1.02	-1100.0	2.0	5
...	...	...	...	...	...	...	...	...
29115	C	26	Value_995#	20.0	1.02	-1100.0	2.0	5
29116	C	26	Value_996#	20.0	1.02	-1100.0	2.0	5
29117	C	26	Value_997#	20.0	1.02	-1100.0	2.0	5
29118	C	26	Value_998#	20.0	1.02	-1100.0	2.0	5
29119	C	26	Value_999#	20.0	1.02	-1100.0	2.0	5

[1120 rows x 8 columns]

So for the following config (Colour=C, Chip=26, V=20, f_R=1.02, dt2=-1100, coverage=2.0), we only have 5 sheets instead of 6. Is it worth dropping this from the dataset?. We're only losing about $\sim 1/30$ th of the data but I don't know if it's too harsh of a threshold.

To further explore the dataset, I need to understand the [physical layout](#).